

Topic 2B: Learning Sequences from Finite Data

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Acknowledgements: These notes use the classical Hellinger analysis of maximum likelihood [1, 2] and its application to sequence learning by Foster, Block, and Misra [3].

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The previous lecture showed that the source distribution minimizes next-token loss when we average over documents drawn from the source. A finite training corpus, however, may leave out the documents on which a competing model incurs large losses. That model can then fit the corpus better than the source, even though its population loss is larger. We ask how much data is needed for the model chosen by training to generate from a distribution close to the source.

1 Learning from a Finite Corpus

Let $\mathcal{V}$ be a finite vocabulary and T the document length, so $\mathcal{U} = \mathcal{V}^T$ is the set of possible documents. The source distribution p assigns probabilities to these documents. Write $x_{<t} = (x_1, \dots, x_{t-1})$ for the prefix before position t . Sampling tokens sequentially from a next-token predictor q gives each complete document the probability $q(x) = \prod_{t=1}^T q(x_t \mid x_{<t})$. We use q to denote both the predictor and the document distribution it induces.

We assume that the model class $\mathcal{Q}$ is finite and contains a model whose document distribution is exactly p . This is the *realizability* assumption. The learner knows the class but does not know which candidate equals p .

Given n independent training documents $X^{(1)}, \dots, X^{(n)} \sim p$, the empirical loss is

$$\hat{L}_{\text{seq}}(q) = \frac{1}{n} \sum_{i=1}^n \sum_{t=1}^T -\log q(X_t^{(i)} \mid X_{<t}^{(i)}) = \frac{1}{n} \sum_{i=1}^n -\log q(X^{(i)}). \quad (1)$$

The loss averages over the n documents and sums over the T token positions. By the chain rule, each document's sum is its negative log probability, so minimizing this loss maximizes the likelihood of the corpus. We write $\hat{q} \in \arg \min_{q \in \mathcal{Q}} \hat{L}_{\text{seq}}(q)$ for the resulting *maximum-likelihood estimator* (MLE).

We measure the accuracy of the generated distribution using *total variation* (TV) distance:

$$\text{TV}(p, q) = \frac{1}{2} \sum_{x \in \mathcal{U}} |p(x) - q(x)| = \max_{B \subseteq \mathcal{U}} |p(B) - q(B)|.$$

Here B is a set of documents representing a property, and $p(B) = \sum_{x \in B} p(x)$ is its probability under the source. Thus $\text{TV}(p, q) \leq \varepsilon$ means that every document property has probability differing by at most ε between the two distributions. We want to know how many training documents are needed for the learned model to achieve this accuracy with high probability.

2 Why a KL Bound Can Fail

The population loss $L_{\text{seq}}(q) = \mathbb{E}_{X \sim p}[-\log q(X)]$ is the expected loss on a fresh source document, whereas $\hat{L}_{\text{seq}}(q)$ averages over the observed corpus. Recall from the previous lecture that the excess population loss is

$$L_{\text{seq}}(q) - L_{\text{seq}}(p) = \sum_{x:p(x)>0} p(x) \log \frac{p(x)}{q(x)} = \text{KL}(p\|q),$$

where $\text{KL}(p\|q)$ is the *Kullback–Leibler* (KL) divergence. It is infinite if q assigns zero probability to any document with positive probability under p . To bound the excess population loss, we might try to show that training and population losses are close for every candidate. Minimizing training loss would then give nearly optimal population loss, and hence small KL divergence. For bounded losses, a concentration inequality and a union bound give exactly this argument.

Concentration and a union bound. Suppose $\ell(q, X) \in [0, M]$, and write $L(q) = \mathbb{E}[\ell(q, X)]$ and $\hat{L}(q) = n^{-1} \sum_{i=1}^n \ell(q, X^{(i)})$. For a fixed model, the empirical loss is an average of independent bounded observations. Hoeffding’s inequality bounds the probability that this average differs substantially from its mean:

$$\mathbb{P}\left[|\hat{L}(q) - L(q)| > \varepsilon\right] \leq 2e^{-2n\varepsilon^2/M^2}.$$

Because training can select any candidate, we need a guarantee that holds for all of them. The union bound controls the probability of *any* candidate having a large deviation by adding the probabilities of the individual deviation events:

$$\mathbb{P}\left[\exists q \in \mathcal{Q} : |\hat{L}(q) - L(q)| > \varepsilon\right] \leq 2|\mathcal{Q}|e^{-2n\varepsilon^2/M^2}.$$

When this quantity is small, with high probability the empirical and population losses are close for all candidates at once. On this event, an empirical minimizer $\hat{q}$ satisfies

$$L(\hat{q}) \leq \hat{L}(\hat{q}) + \varepsilon \leq \hat{L}(p) + \varepsilon \leq L(p) + 2\varepsilon.$$

The middle inequality uses empirical minimization; the other two use the simultaneous bounds.

Hoeffding’s inequality requires a common upper bound on the loss of every document under every candidate. We have no such bound for log loss: a candidate can assign a document an arbitrarily small probability, making its loss arbitrarily large. If that document is rare under the source, the training sample may miss it entirely, even when it contributes substantially to population loss.

Example. Suppose the source produces only the two documents below, where $0 < \rho \leq 1/2$. We compare it with a model q_{cats} that always produces *Cats purr.*, so the model class is $\mathcal{Q} = \{p, q_{\text{cats}}\}$.

Document	Source p	q_{cats}
<i>Cats purr.</i>	$1 - \rho$	1
<i>Dogs bark.</i>	ρ	0

Both distributions assign zero probability to all other documents. They can be implemented by next-token predictors that differ only in their first-token prediction: after *Cats*, both predict *purr.*, and after *Dogs*, both predict *bark.*

On a corpus consisting entirely of *Cats purr.*, the model q_{cats} has training loss zero, while p has training loss $-\log(1 - \rho) > 0$, so training selects q_{cats} . If *Dogs bark.* appears even once, q_{cats} has infinite training loss and training selects p .

The corpus contains only *Cats purr.* with probability $(1 - \rho)^n$. On this event, the learned model satisfies

$$\text{KL}(p \parallel \hat{q}) = \infty, \quad \text{TV}(p, \hat{q}) = \rho.$$

The KL divergence is infinite because the learned model assigns zero probability to a document that the source can produce. When ρ is small, that document can easily be absent from the corpus.¹ The effect on TV, however, is only the mass ρ moved from *Dogs bark.* to *Cats purr.* The generated distribution can therefore be close to the source in TV even when its KL divergence is infinite.

3 Learning the Generated Distribution

Although empirical loss minimization can fail to give a finite KL bound, it does give a TV guarantee for any finite class containing p .

Theorem 3.1 (Finite-class MLE; TV consequence of [3, Proposition 2.1]). *Let $\mathcal{Q}$ be finite and contain the source distribution p . Let $\hat{q}$ minimize empirical next-token log loss on n independent documents from p . For $\varepsilon, \delta \in (0, 1)$, if*

$$n \geq \frac{2 \log(|\mathcal{Q}|/\delta)}{\varepsilon^2}, \tag{2}$$

then $\text{TV}(p, \hat{q}) \leq \varepsilon$ with probability at least $1 - \delta$ over the training sample.

We choose an accuracy ε and a failure probability δ before drawing the training corpus. With the stated number of documents, the learned model satisfies $|p(B) - \hat{q}(B)| \leq \varepsilon$ for every document property $B \subseteq \mathcal{U}$, with probability at least $1 - \delta$ over the corpus. For example, the probability that a fresh generation has matching opening and closing tags is within ε of the corresponding source probability. The error ε concerns generation probabilities; the failure probability δ concerns the training corpus. For each corpus where the TV bound holds, it controls all document properties simultaneously, so no additional union bound over properties is needed.

The sample size n counts independent documents. Tokens within a document can depend on earlier tokens, as described by the conditionals in $q(x) = \prod_t q(x_t \mid x_{<t})$, so a corpus of n length- T documents need not provide nT independent observations. In the proof, we use independence only between documents.

3.1 Proof of Theorem 3.1

We need to bound the chance that training selects a model with TV distance greater than ε from p . Since p belongs to the class, a candidate can be selected only if its training loss is no larger than p 's. Thus, for each fixed $q \in \mathcal{Q}$,

$$\{\hat{q} = q\} \subseteq \{\hat{L}_{\text{seq}}(q) \leq \hat{L}_{\text{seq}}(p)\}.$$

We will show that a model far from p in TV is unlikely to fit a sample from p at least as well as p itself. Specifically, when $\text{TV}(p, q) > \varepsilon$, the event on the right has probability at most $e^{-n\varepsilon^2/2}$. A union bound over these candidates will then bound the chance of selecting a model with TV error greater than ε .

¹If $n\rho \leq 1/2$, then $(1 - \rho)^n \geq 1 - n\rho \geq 1/2$. Since ρ can be arbitrarily small, no finite KL bound depending only on n , $|\mathcal{Q}|$, and confidence can hold uniformly over these examples. For each fixed $\rho > 0$, the probability of missing the document tends to zero as $n \rightarrow \infty$.

Fix $q \in \mathcal{Q}$, and write $D = (x^{(1)}, \dots, x^{(n)}) \in \mathcal{U}^n$ for an ordered training sample of n complete documents. The two models assign this corpus likelihoods

$$P_p(D) = \prod_{i=1}^n p(x^{(i)}), \quad P_q(D) = \prod_{i=1}^n q(x^{(i)}).$$

Because the sample is drawn from p , its actual probability is $P_p(D)$. By (1), the candidate has training loss no larger than p 's exactly when $P_q(D) \geq P_p(D)$. Thus the necessary condition for selecting q has probability

$$\mathbb{P}[\widehat{L}_{\text{seq}}(q) \leq \widehat{L}_{\text{seq}}(p)] = \sum_{D: P_q(D) \geq P_p(D)} P_p(D). \quad (3)$$

The right-hand side adds the probabilities, under the source, of all training corpora on which q fits at least as well as p .

For every corpus included in the sum, we have $P_p(D) \leq P_q(D)$. Replacing each term by this larger quantity gives a valid inequality, but the resulting bound may be as large as one:

$$\sum_{D: P_q(D) \geq P_p(D)} P_p(D) \leq \sum_{D: P_q(D) \geq P_p(D)} P_q(D) \leq \sum_{D \in \mathcal{U}^n} P_q(D) = 1.$$

To get a more useful bound, we use the geometric mean, which lies between the two likelihoods on these corpora:

$$P_p(D) \leq \sqrt{P_p(D)P_q(D)} \leq P_q(D) \quad \text{when } P_q(D) \geq P_p(D).$$

Substituting this upper bound into (3) gives

$$\begin{aligned} \mathbb{P}[\widehat{L}_{\text{seq}}(q) \leq \widehat{L}_{\text{seq}}(p)] &\leq \sum_{D: P_q(D) \geq P_p(D)} \sqrt{P_p(D)P_q(D)} \\ &\leq \sum_{D \in \mathcal{U}^n} \sqrt{P_p(D)P_q(D)}. \end{aligned}$$

The second inequality adds the remaining corpora, whose contributions are nonnegative. The resulting sum retains both P_p and P_q . Because the documents are independent, it factors as

$$\begin{aligned} \sum_{D \in \mathcal{U}^n} \sqrt{P_p(D)P_q(D)} &= \sum_{x^{(1)}, \dots, x^{(n)}} \prod_{i=1}^n \sqrt{p(x^{(i)})q(x^{(i)})} \\ &= \left(\sum_{x \in \mathcal{U}} \sqrt{p(x)q(x)} \right)^n. \end{aligned} \quad (4)$$

Expanding the n -th power gives one term for each ordered training sample, which explains the final equality. The factor $\sum_x \sqrt{p(x)q(x)}$ measures the overlap between the two distributions.

We relate this overlap to TV through the *Hellinger distance*, the Euclidean distance between their square-root probability vectors [4, Chapters 3–4]. Expanding its square gives

$$d_{\text{H}}^2(p, q) := \sum_x (\sqrt{p(x)} - \sqrt{q(x)})^2 = 2 \left(1 - \sum_x \sqrt{p(x)q(x)} \right). \quad (5)$$

We will bound $d_H(p, \hat{q})$ and then use $\text{TV}(p, q) \leq d_H(p, q)$ to obtain the TV guarantee. Hellinger distance measures the error of the model trained by minimizing log loss.

The identity above shows that the overlap is at most one, with equality only when $p = q$. For any other candidate it is strictly below one, so its n -th power decreases exponentially as the number of independent documents grows.

Substituting $1 - d_H^2(p, q)/2$ for the overlap in (4) and using $(1 - u)^n \leq e^{-nu}$, we obtain

$$\mathbb{P}[\hat{L}_{\text{seq}}(q) \leq \hat{L}_{\text{seq}}(p)] \leq e^{-nd_H^2(p, q)/2}. \quad (6)$$

Thus, whenever $d_H(p, q) > \varepsilon$, the chance that q fits the sample at least as well as p is at most $e^{-n\varepsilon^2/2}$.

Training can select any candidate in the class. If $d_H(p, \hat{q}) > \varepsilon$, then at least one candidate at Hellinger distance greater than ε from p has training loss no larger than p 's. Taking a union bound over all such candidates therefore gives

$$\begin{aligned} \mathbb{P}[d_H(p, \hat{q}) > \varepsilon] &\leq \sum_{\substack{q \in \mathcal{Q}: \\ d_H(p, q) > \varepsilon}} \mathbb{P}[\hat{L}_{\text{seq}}(q) \leq \hat{L}_{\text{seq}}(p)] \\ &\leq |\mathcal{Q}| e^{-n\varepsilon^2/2} \leq \delta. \end{aligned}$$

There are at most $|\mathcal{Q}|$ terms, and the sample size in (2) makes their sum at most δ . Thus $d_H(p, \hat{q}) \leq \varepsilon$ with probability at least $1 - \delta$.

To finish, we show that $\text{TV}(p, q) \leq d_H(p, q)$ by factoring $p(x) - q(x)$ into square-root terms and applying Cauchy–Schwarz:

$$\begin{aligned} \text{TV}(p, q) &= \frac{1}{2} \sum_x |p(x) - q(x)| \\ &= \frac{1}{2} \sum_x \left| \sqrt{p(x)} - \sqrt{q(x)} \right| \left(\sqrt{p(x)} + \sqrt{q(x)} \right) \\ &\leq \frac{1}{2} d_H(p, q) \left(\sum_x (\sqrt{p(x)} + \sqrt{q(x)})^2 \right)^{1/2} \leq d_H(p, q). \end{aligned}$$

The first factor is Hellinger distance by definition. The remaining square root is at most two: applying $(a + b)^2 \leq 2a^2 + 2b^2$ to each term bounds the sum inside it by $2 \sum_x p(x) + 2 \sum_x q(x) = 4$. Therefore, whenever $d_H(p, \hat{q}) \leq \varepsilon$, we also have $\text{TV}(p, \hat{q}) \leq \varepsilon$, which proves the theorem.

3.2 Next-Token Prediction and Document Length

Each training document gives an example at every token position, but only for the prefixes it contains. Many possible prefixes may appear rarely or never, so we cannot expect to estimate the conditional distribution accurately at every prefix. The theorem allows these errors while controlling the distribution of generated documents. As in the previous lecture, summing the next-token log losses gives the negative log probability of the document. The likelihood used in training is therefore exactly the probability of generating that document by sampling successive tokens from the predictor. The TV guarantee applies when the model conditions on its own generated prefixes, even though training uses the observed prefixes.

The proof compares complete predictors: for each candidate whose document distribution is far from p , it bounds the chance of fitting the corpus at least as well as p , then takes a union bound

over the class. For a fixed class $\mathcal{Q}$ containing p , the sufficient number of documents depends on $\log |\mathcal{Q}|$, with no additional factor of T . These n documents still contain nT tokens, and tokens within a document need not be independent.

Document length can affect the size of the class. If we choose among M predictors separately at each position, there can be M^T complete models, so $\log |\mathcal{Q}| = T \log M$. If we instead choose one of M predictors and use it at every position, there are at most M complete models. Each predictor is a function of the prefix, so using the same predictor throughout still allows its next-token distribution to change with the preceding text. The theorem applies to either choice of class, provided that it contains p [3, Section 2.3].

A neural model class is usually infinite, so we need a measure of complexity other than its cardinality. If a finite collection of representatives approximates every model’s log likelihood to the required accuracy, we can take the union bound over those representatives. The size of such a collection, a *covering number*, replaces the finite class size, with an additional term for approximation error.

References

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